

WASKISH, MN, USA

WMO: 727486

Lat: 48.154N

Lon: 94.517W

Elev: 360

StdP: 97.07

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04983

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-27.7	-33.8	0.2	-31.6	-31.1	0.2	-27.4	8.8	-9.6	8.1	-9.7	1.7	240	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	29.0	22.3	27.6	20.9	26.3	19.7	23.5	27.9	22.2	26.4	21.0	25.2	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.6	27.0	21.0	16.4	25.7	19.1	14.5	23.7	71.7	27.7	66.7	25.9	62.5	25.5	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.8	7.0	DB	-34.6	32.8	3.6	2.9	-37.2	34.8	-39.3	36.5	-41.3	38.1	-43.9	40.2	
			WB	-35.1	25.2	3.6	1.1	-37.7	25.9	-39.8	26.5	-41.9	27.1	-44.5	27.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-13.5	-12.3	-3.8	4.2	11.1	16.9	19.6	18.1	13.9	6.3	-1.8	-9.9
	DBStd	12.99	8.47	7.50	7.76	5.50	4.84	3.50	3.11	3.37	4.63	4.90	6.20	7.08
	HDD10.0	3136	728	623	429	184	44	1	0	0	15	138	354	618
	HDD18.3	5319	986	857	686	424	229	67	23	46	147	375	603	877
	CDD10.0	1002	0	0	2	10	78	207	298	252	131	23	1	0
	CDD18.3	144	0	0	0	0	5	23	63	39	13	1	0	0
	CDH23.3	1262	0	0	0	3	71	207	522	338	109	12	0	0
	CDH26.7	246	0	0	0	0	14	33	109	68	19	2	0	0
Wind	WSAvg	2.9	2.8	2.8	2.9	3.2	3.2	2.8	2.6	2.5	2.9	3.2	3.2	2.8
Precipitation	PrecAvg	656	19	16	28	41	76	108	99	85	74	59	32	23
	PrecMax	797	45	30	77	91	158	276	167	171	175	111	86	47
	PrecMin	457	6	1	8	17	44	36	37	32	15	7	4	8
	PrecStd	103	9	8	17	16	34	53	34	36	40	28	21	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	7.3	18.8	22.6	28.9	29.8	31.4	31.2	28.8	25.1	16.1	5.9
		MCWB	2.3	4.1	13.1	14.0	19.5	21.6	24.0	23.4	22.3	16.0	10.7	3.5
	2%	DB	2.3	2.9	12.7	18.9	25.2	27.4	29.0	28.0	27.0	20.9	12.3	2.8
		MCWB	1.2	1.1	8.4	11.3	16.4	20.2	22.8	21.6	20.2	14.5	9.2	1.5
	5%	DB	0.9	1.3	8.8	16.3	22.6	26.0	27.6	27.1	23.9	17.2	8.9	1.2
		MCWB	0.0	0.2	5.5	9.2	15.0	19.0	21.5	20.8	18.8	12.4	6.4	0.5
	10%	DB	-2.0	-1.0	6.1	13.7	20.3	23.9	26.3	25.2	22.2	13.8	7.0	-0.3
		MCWB	-2.6	-1.7	3.4	7.9	14.0	18.1	20.2	19.5	17.8	10.2	4.9	-0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	4.3	14.8	14.7	20.5	23.8	25.1	24.1	23.1	17.2	12.3	3.8
		MCDB	4.0	7.4	17.2	20.8	26.1	28.0	29.4	29.0	27.8	21.6	14.2	4.9
	2%	WB	1.1	1.7	8.6	11.8	18.2	21.7	23.7	22.9	21.2	15.1	8.9	1.8
		MCDB	2.0	3.1	12.9	17.3	23.1	26.0	28.1	27.3	25.1	20.3	11.6	2.8
	5%	WB	-0.5	0.1	5.7	10.1	16.4	20.1	22.5	21.6	19.6	12.6	6.8	0.4
		MCDB	0.4	1.4	8.6	15.4	21.0	24.2	26.6	25.7	23.1	15.8	8.6	1.1
	10%	WB	-2.2	-1.8	3.5	8.5	14.8	19.0	21.3	20.4	18.1	10.9	4.7	-1.0
		MCDB	-1.9	-1.1	5.8	13.5	19.1	22.9	25.4	24.4	21.0	13.6	6.9	-0.5
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.6	11.8	11.8	11.9	11.0	11.2	11.7	11.1	9.1	7.7	8.4
		MCDBR	8.9	11.0	13.7	16.4	15.9	13.3	12.4	13.3	13.6	13.9	11.1	8.7
		MCWBR	7.8	8.9	9.2	9.9	8.7	7.4	7.3	7.6	8.3	8.6	7.8	7.6
	5% WB	MCDBR	8.4	10.1	12.4	14.8	13.3	11.2	11.4	11.8	11.6	11.9	10.2	8.3
		MCWBR	7.4	8.3	8.7	9.5	8.2	7.0	7.4	7.2	7.8	7.8	7.6	7.4
Clear-Sky Solar Irradiance	taub	0.254	0.277	0.306	0.352	0.363	0.375	0.382	0.377	0.350	0.306	0.279	0.257	
	taud	2.361	2.306	2.388	2.366	2.392	2.381	2.382	2.418	2.473	2.576	2.510	2.397	
	Ebn at Noon	836	887	915	899	899	887	874	864	857	854	805	785	
	Edh at Noon	67	90	99	112	114	116	115	106	91	70	59	58	
All-Sky Solar Radiation	RadAvg	1.48	2.67	3.85	4.81	5.36	5.71	6.13	5.14	3.79	2.27	1.32	1.06	
	RadStd	0.16	0.23	0.39	0.42	0.32	0.43	0.28	0.38	0.29	0.34	0.14	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page